Mathematical Equations of Sensory Processing

Chapter 6: Quantitative Models of Perception and Sensation

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Abstract

This document presents mathematical models for all major sensory systems. Each modality — vision, audition, somatosensation, olfaction, gustation, and vestibulation — is quantified using published computational equations. These models span from peripheral transduction mechanisms to cortical feature extraction, covering psychophysical laws, neural tuning functions, and multisensory integration. Six case studies demonstrate practical application of these models to sensory perception problems.

Contents

1 General Sensory Transduction Principles

1.1 Weber-Fechner Law: Threshold Sensitivity

$$S = k \ln\left(\frac{I}{I_0}\right) \tag{1}$$

Variable	Definition	Units
S	Perceived sensation magnitude	dimensionless
k	Psychophysical scaling constant	varies by modality
I	Stimulus intensity	varies by modality
I_0	Threshold intensity	varies by modality

Explanation: The Weber fraction $\Delta I/I = c$ predicts the relationship between just-noticeable difference and stimulus intensity. The relationship holds approximately across vision, audition, touch, and taste over mid-range intensities.
Key Reference: Fechner, G. T. (1860). *Elemente der Psychophysik.* Breitkopf und Härtel.

1.2 Stevens' Power Law: Perceived Magnitude Scaling

$$S = k \cdot I^n \tag{2}$$

Variable	Definition / Typical Range
n	Power exponent (0.3 to 3.5)

Explanation: Vision brightness $n \approx 0.33$, sound loudness $n \approx 0.67$, pain pressure $n \approx 1.1$. When $n < 1$, sensation is compressive; when $n > 1$, sensation is expansive.
Key Reference: Stevens, S. S. (1957). On the psychophysical law. *Psychological Review*, 64(3), 153–181.

1.3 Sensory Adaptation: Exponential Decay

$$R(t) = R_0 e^{-t/\tau_a} + R_{ss} \tag{3}$$

Variable	Definition	Typical Value
τ_a	Adaptation time constant	50–500 ms

Explanation: Rapidly adapting (RA) sensors: $\tau_a \approx$ 10-50 ms. Slowly adapting (SA) sensors: $\tau_a \approx$ 500-2000 ms. Adaptation suppresses sustained background signals.
Key Reference: Laughlin, S. B. (1989). The role of sensory adaptation in the retina. *Physica D*, 38(1–3), 1–27.

1.4 Naka-Rushton Contrast Response

$$\frac{R}{R_{\max}} = \frac{C^n}{C_{50}^n + C^n} \tag{4}$$

Variable	Definition	Typical Value
C_{50}	Half-saturation contrast	10–50%
n	Exponent (slope)	1.5–3.0

Explanation: This equation fits contrast-response relationships in photoreceptors, retinal ganglion cells, and cortical neurons. The exponent $n > 1$ reflects sigmoidal saturation.
Key Reference: Naka, K. I., & Rushton, W. A. (1966). S-potentials from luminosity units in the retina of fish. *Journal of Physiology*, 185(3), 587–599.

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2 Visual Processing

2.1 Difference-of-Gaussians (DoG) Center-Surround RF

$$w(r) = A_e \exp\left(-\frac{r^2}{2\sigma_e^2}\right) - A_i \exp\left(-\frac{r^2}{2\sigma_i^2}\right) \tag{5}$$

Variable	Definition	Typical Value
σ_e	Center Gaussian SD	0.1–1.0
σ_i	Surround Gaussian SD	2–5 $\times$ σ_e

Explanation: DoG RFs emerge from horizontal cell lateral inhibition. Retinal ganglion cells (midget M, parasol P, bistratified K) all exhibit DoG tuning. Surround suppression enhances edge contrast.
Key Reference: Rodieck, R. W. (1965). Quantitative analysis of cat retinal ganglion cell response to visual stimuli. *Vision Research*, 5(12), 583–601.

2.2 Gabor Filter: V1 Orientation Selectivity

$$G(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos(2\pi f x' + \phi) \tag{6}$$

Variable	Definition	Typical Value
f	Spatial frequency	0.5–4.0 cpd
γ	Aspect ratio	0.3–0.7

Explanation: Gabor filters predict V1 simple cell responses. Orientation tuning arises from elongated thalamocortical inputs. Spatial frequency preference matches input statistics.
Key Reference: Jones, J. P., & Palmer, L. A. (1987). An evaluation of the two-dimensional Gabor filter model of simple receptive fields in cat striate cortex. *Journal of Neurophysiology*, 58(6), 1233–1258.

2.3 Motion Energy Model: V5/MT Direction Selectivity

$$E(x, y, t) = [L_e(x, y, t)]^2 + [L_o(x, y, t)]^2 \tag{7}$$

Explanation: Even and odd filters offset in phase. Energy is invariant to contrast and responds to preferred-direction motion. Predicts MT tuning for speed, direction, and disparity.
Key Reference: Adelson, E. H., & Bergen, J. R. (1985). Spatiotemporal energy models for the perception of motion. *Journal of the Optical Society of America A*, 2(2), 284–299.

2.4 Phototransduction: Single-Photon Response

$$F(t) = 1 - \exp\left(-\frac{1}{2}A\Phi t_{\text{eff}}^2\right) \tag{8}$$

Variable	Definition	Typical Value
A	Amplification constant	4–8 $(\text{msec})^{-2}$

Explanation: Single photon activates $\sim$ 100 transducin molecules. cGMP hydrolysis blocks CNG channels, hyperpolarizing the cell. Response time constant $\sim$ 200 ms in rods; $\sim$ 10$\times$ faster in cones.
Key Reference: Lamb, T. D., & Pugh, E. N. (1992). A quantitative account of the activation steps involved in phototransduction in amphibian photoreceptors. *Journal of Physiology*, 449(1), 719–758.

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3 Auditory Processing

3.1 Greenwood Tonotopy Function: Cochlear Frequency Map

$$f(X) = A\left(10^{aX} - k\right) \tag{9}$$

Parameter	Definition	Typical Value
A	Scaling constant	165 Hz
a	Frequency scaling constant	2.1
k	Offset constant	0.88

Explanation: High frequencies code near the base; low frequencies near the apex. For human cochlea (L=35 mm): X=0 $\rightarrow$ f$\approx$ 20 kHz; X=1 $\rightarrow$ f$\approx$ 20 Hz. Exponential mapping compresses low frequencies.
Key Reference: Greenwood, D. D. (1961). Critical bandwidth and the frequency coordinates of the basilar membrane. *Journal of the Acoustical Society of America*, 33(10), 1344–1356.

3.2 Gammatone Filter: A1 Frequency Tuning

$$g(t) = t^{n-1} e^{-2\pi bt} \cos(2\pi f_c t + \phi) \tag{10}$$

Variable	Definition	Typical Value
n	Filter order	4
b	Bandwidth parameter	25–200 Hz
f_c	Center frequency	50–16000 Hz

Explanation: Approximates cascade of first-order filters. Bandwidth b increases with f_c (constant-Q ≈ 10). Four-fold symmetry fits empirical auditory nerve responses.
Key Reference: Patterson, R. D., Nimmo-Smith, I., Weber, D. L., & Milroy, R. (1988). The deterioration of hearing with age: Frequency selectivity, the absolute threshold, and the stapedius reflex. *Journal of the Acoustical Society of America*, 72(6), 1549–1559.

3.3 Jeffress ITD Coincidence Detection: Binaural Localization

$$r(\tau) = \int_{-\infty}^{\infty} r_L(t) \cdot r_R(t+\tau)\, dt \tag{11}$$

Explanation: Medial superior olive (MSO) receives left/right inputs with tuned axonal delays. Peak cross-correlation occurs when τ matches head arrival time. Humans resolve ITDs to ~ 10 μsec ($\approx$ 0.3azimuth).
Key Reference: Jeffress, L. A. (1948). A place theory of sound localization. *Journal of Comparative and Physiological Psychology*, 41(1), 35–39.

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4 Multisensory Integration

4.1 Maximum Likelihood Estimation (MLE): Optimal Cue Combination

$$\hat{s}_{\text{MLE}} = \frac{\hat{s}_1/\sigma_1^2 + \hat{s}_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2} \tag{12}$$

Explanation: Each cue weighted inversely by variance (reliability). More reliable cues receive higher weight. Posterior parietal cortex, superior colliculus, and insular cortex integrate multisensory estimates.
Key Reference: Ernst, M. O., & Banks, M. S. (2002). Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415(6870), 429–433.

4.2 Combined Variance in MLE

$$\sigma_{\text{MLE}}^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} \tag{13}$$

Explanation: If $\sigma_1 = 3$ (vision) and $\sigma_2 = 6$ (proprioception), then $\sigma_{\text{MLE}} = 2.68$ — better than vision alone. Neurons in PPC and SC exhibit cue-combination properties consistent with inverse-variance weighting.

Key Reference: Ernst, M. O., & Banks, M. S. (2002). Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415(6870), 429–433.

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5 Somatosensation and Pain

5.1 Gate Control Theory of Pain Modulation

$$T = R \cdot [L(1 + aA) - I(bA + c)] \tag{14}$$

Variable	Definition	Typical Value
a, b, c	Weighting coefficients	$\sim$ 0.5–1.0

Explanation: A-beta activation closes gate (blocks pain); C-fiber activation opens it. Serotonergic input from nucleus raphe magnus enhances inhibition. Enkephalin-releasing dorsal horn interneurons suppress pain transmission.
Key Reference: Melzack, R., & Wall, P. D. (1965). Pain mechanisms: A new theory. *Science*, 150(3699), 971–979.

5.2 Sensory Adaptation in Touch: SA and RA Neurons

$$R(t) = R_\infty + (R_0 - R_\infty)e^{-t/\tau_{\text{adapt}}} \tag{15}$$

Variable	Definition	Typical Value
τ_{adapt}	Adaptation time constant	10–2000 ms

Explanation: RA neurons: $\tau_{\text{adapt}} \approx$ 10-100 ms, $R_\infty \approx 0$ (phasic). SA neurons: $\tau_{\text{adapt}} \approx$ 500-2000 ms, $R_\infty > 0.3R_0$ (tonic). Merkel cells maintain sustained firing. Ruffini endings sustain response to skin stretch.
Key Reference: Freeman, A. W., & Johnson, K. O. (1982). A model accounting for effects of vibratory amplitude and frequency on the responses of cutaneous mechanoreceptors. *Journal of Neurophysiology*, 48(6), 1601–1626.

5.3 Nociceptor Sensitization: TRPV1 Activation

$$P_{\text{open}} = \frac{[Cap]^n}{K_d^n + [Cap]^n} \tag{16}$$

Variable	Definition	Typical Value
K_d	Half-maximal concentration (EC50)	100–500 nM
n	Hill coefficient	1.2–1.5

Explanation: TRPV1 activated by capsaicin, anandamide, and heat (> 43°C). Inflammation releases bradykinin and prostaglandins, sensitizing TRPV1. TRPM8 (menthol, cold < 25°C) and TRPA1 (irritants) use similar mechanisms.
Key Reference: Caterina, M. J., Schumacher, M. A., Tominaga, M., Rosen, T. A., Levine, J. D., & Julius, D. (1997). The capsaicin receptor: A heat-activated ion channel in the pain pathway. *Nature*, 389(6653), 816–824.

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6 Olfaction

6.1 Combinatorial Coding: Olfactory Receptor Capacity

$$N_{\text{discriminable}} = \binom{n_{\text{OR}}}{k} = \frac{n_{\text{OR}}!}{k!(n_{\text{OR}} - k)!} \tag{17}$$

Variable	Definition
n_{OR}	Total number of ORs (400 humans)
k	Number of active ORs per odor (5–20 typical)

Explanation: With $n_{\text{OR}} = 400$ and average $k = 10$, $N \approx 6.7 \times 10^{15}$. Humans can discriminate > 1 trillion distinct odors. Combinatorial coding compresses large odorant space into ~ 400 receptor responses via nonlinear integration in piriform cortex.
Key Reference: Cleland, T. A., & Linster, C. (2005). Computation in the olfactory system. *Neural Computation*, 17(8), 1729–1755.

6.2 Piriform Cortex Pattern Completion: Hopfield Network

$$E = -\frac{1}{2} \sum_{i \neq j} w_{ij} s_i s_j \tag{18}$$

Explanation: Hopfield networks trained by Hebbian rule. Activity evolves to minimize E, converging to stored pattern closest to input (pattern completion). Piriform capacity $P_{\max} \approx 0.138N$. Sparse codes increase capacity.
Key Reference: Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8), 2554–2558.

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7 Gustation (Taste)

7.1 Taste Receptor Activation: Hill Equation for GPCR

$$R_{\text{active}} = R_{\max} \frac{[L]^n}{K_d^n + [L]^n} \tag{19}$$

Variable	Definition	Typical Value
K_d	EC50	0.1–10 mM
n	Hill coefficient	1.0–4.0

Explanation: Sweet: T1R2+T1R3 binds sugars ($K_d \approx$ 1-10 mM), activates Gαgustducin $\rightarrow$ PLCβ2 $\rightarrow$ IP3 $\rightarrow$ Ca^{2+} $\rightarrow$ TRPM5. Umami: T1R1+T1R3 binds L-glutamate. Bitter: $\sim$ 25 T2Rs, $K_d \approx$ 10-100 μM.
Key Reference: Breslin, P. A., & Huang, L. (2006). Human taste: Peripheral anatomy, taste transduction, and perception. *Advances in Oto-Rhino-Laryngology*, 63, 152–190.

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8 Vestibular System

8.1 Vestibulo-Ocular Reflex (VOR) Gain

$$G_{\text{VOR}} = \frac{\Delta\theta_{\text{eye}}}{\Delta\theta_{\text{head}}} \tag{20}$$

Value	Definition
G_{VOR}	VOR gain (0.95–1.05 typical)

Explanation: VOR is a three-neuron arc with $\sim$ 10 ms latency. Gain $>$ 1.0 causes overshooting; gain $<$ 1.0 causes undershoot. Cerebellum (flocculonodular lobe) calibrates gain. Vestibular nucleus or cerebellar damage causes nystagmus.
Key Reference: Robinson, D. A. (1981). The use of control systems analysis in the neurophysiology of eye movements. *Annual Review of Neuroscience*, 4, 463–503.

8.2 Semicircular Canal Transfer Function

$$H(s) = \frac{s\,T_1}{sT_1 + 1} \cdot \frac{sT_2}{sT_2 + 1} \tag{21}$$

Variable	Definition	Typical Value
T_1	Long time constant (velocity storage)	15–80 s
T_2	Short time constant (cupula restoring)	0.003–0.005 s

Explanation: At low frequencies (DC), output decays with $T_1 \approx$ 80 s. Velocity storage integrator extends T_1 to $\sim$ 20 s. At high frequencies, cupula inertia dominates with $T_2 \approx$ 3 ms. Bandpass response from 0.01–10 Hz.
Key Reference: Mayne, R. (1974). A systems concept of the vestibular organs. In H. H. Kornhuber (Ed.), *Handbook of sensory physiology* (Vol. 6, pp. 493–580). Springer.

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9 Proprioception and Interoception

9.1 Muscle Spindle Ia Afferent Firing Rate

$$f_{Ia}(t) = k_d \frac{dL}{dt} + k_s L(t) + f_0 \tag{22}$$

Variable	Definition	Typical Value
k_d	Dynamic sensitivity (velocity gain)	0.5–5.0 Hz/(mm/s)
k_s	Static sensitivity (length gain)	0.5–3.0 Hz/mm
f_0	Baseline firing rate	5–20 spike/s

Explanation: Ia fibers wrap around intrafusal nuclear bag fibers (dynamic) and nuclear chain fibers (static). Bag fibers contribute k_d; chain fibers contribute k_s. Gamma motor neurons adjust both gains during voluntary contraction (alpha-gamma coactivation).
Key Reference: Matthews, P. B. (1964). Muscle spindles and their motor control. *Physiological Reviews*, 44(2), 219–288.

9.2 Interoceptive Predictive Coding and Free Energy

$$F = D_{\mathrm{KL}}[q(\phi) \| p(\phi|s)] - \ln p(s) \tag{23}$$

Explanation: Brain maintains generative model predicting interoceptive inputs. Prediction errors trigger model updates, minimizing free energy. Anterior insular cortex integrates lamina I interoceptive input with predictions from vmPFC. Interoceptive accuracy (heartbeat, breathing sensing) correlates with emotional awareness and empathy.
Key Reference: Friston, K. J., Stephan, K. E., Montague, R., & Dolan, R. J. (2010). Computational psychiatry: the brain as a phantastic organ of adaptation. *Nature Reviews Neuroscience*, 16(1), 13–25.

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10 Case Studies

10.1 Case Study 1: Stevens' Power Law — Pain Magnitude Estimation

At force $I = 0.1$ kPa: $S_1 = k \times 0.1^{1.1}$

At force $I = 2.0$ kPa: $S_3 = k \times 2.0^{1.1}$

If $S_1 = 10$, then $k = 126$. Check: $S_3 = 126 \times 2.21 = 278.5$ (suggests saturation ceiling).

10.2 Case Study 2: DoG Receptive Field Response

Retinal GC with $\sigma_e = 0.2$, $\sigma_i = 0.6$, $A_e = 1.0$, $A_i = 0.5$. Light spot (2°) centered on RF: Net response $R \approx 0.6$ (normalized). Larger spot recruits more surround inhibition.

10.3 Case Study 3: Greenwood Tonotopy — Speech Frequencies

For $f = 500$ Hz: $X = 0.32 \to 11.2$ mm from apex.
For $f = 2$ kHz: $X = 0.53 \to 18.6$ mm from apex.
For $f = 8$ kHz: $X = 0.80 \to 28$ mm from apex.
Speech frequencies span $\sim 30\%$ of cochlear length.

10.4 Case Study 4: Jeffress ITD Detection — 45Azimuth

Head diameter $D = 0.17$ m, azimuth $\theta = 45$, speed of sound $c = 343$ m/s.

$$\text{ITD} = \frac{0.17 \times 0.707}{343} = 0.35 \text{ ms}$$

MSO neurons tuned to 45° have $\tau_{\text{best}} \approx 0.35$ ms. Human ITD threshold: ~ 10 μs.

10.5 Case Study 5: MLE Cue Combination — Hand Position

Visual: $\hat{s}_{\text{vis}} = 20$ cm, $\sigma_{\text{vis}} = 3$ cm.
Proprioceptive: $\hat{s}_{\text{prop}} = 22$ cm, $\sigma_{\text{prop}} = 6$ cm.

$$\hat{s}_{\text{MLE}} = \frac{20/9 + 22/36}{1/9 + 1/36} = 20.4 \text{ cm}$$

$$\sigma_{\text{MLE}} = \sqrt{7.2} = 2.68 \text{ cm}$$

10.6 Case Study 6: Muscle Spindle Ia Firing Rate During Stretch

Resting length $L_0 = 50$ mm $\to L = 55$ mm over 0.5 s (velocity = 10 mm/s).
$k_d = 2$ Hz/(mm/s), $k_s = 3$ Hz/mm, $f_0 = 15$ spike/s.
Phase 1 — Dynamic: $f_{Ia} = 2 \times 10 + 3 \times 52.5 + 15 = 192.5$ spike/s.
Phase 2 — Static: $f_{Ia} = 0 + 3 \times 55 + 15 = 180$ spike/s.
Phase 3 — Recovery: $f_{Ia} = 2 \times (-10) + 3 \times 50 + 15 = 145$ spike/s.

Next: Chapter 7 — Memory Systems